

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 4: Brovary Optical Calibration

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Overview

The optical coupling is inferred from the Brovary eclipse plates with plate scale and centroid uncertainty carried explicitly.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\delta\theta(r) = \gamma 2GM/(c^2 r)$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$\gamma = 2.00 \pm 0.22$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$c_{read} = \lambda c_{medium}$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper A, §§1–5
- Brovary Plate Reduction Manual r3
- SGT9 optical block O-11–O-19